



IILM University — Internship Evaluation Presentation

Autonomous Multi-Agent Cyber Defense System (AMCDS)

Research Internship at IIT Bombay

Student Details

Name: Kaustubh Pandey

Enrollment: CS-23411064

Program: B. Tech 4th Year

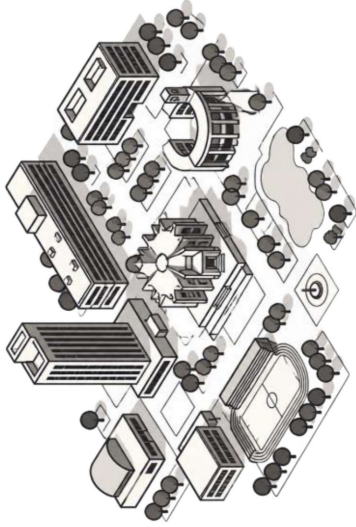
Internship Details

Organization: IIT Bombay, Dept. of
CSE

Supervisor: Prof. K. Sandeep

Duration: 30 May 2026 – 30 July
2026

Research Internship Overview



Internship at a Glance

- **Organization:** Indian Institute of Technology Bombay
- **Department:** Computer Science and Engineering
- **Type:** Research Internship
- **Duration:** 30 May 2026 – 30 July 2026 (2 months)
- **Mode:** Hybrid (Remote and On-Campus)
- **Supervisor:** Prof. K. Sandeep
- **Domain:** AI-Driven Cybersecurity & Network Defense
- **Project:** Autonomous Multi-Agent Cyber Defense System (AMCDS)

Organization Profile — IIT Bombay

Established

1958 — Premier institution for engineering education and research in India

Location

Powai, Mumbai — One of India's most prestigious academic campuses

Research Focus

Computer science, artificial intelligence, cybersecurity, and data engineering

Department Strengths

Machine learning, distributed systems, network security, and advanced computing

IIT Bombay offers a highly stimulating research ecosystem — enabling industry-relevant, cutting-edge work at the intersection of AI and cybersecurity.

Problem Statement

Current Challenges in Network Security

The Gaps

- Sophisticated cyberattacks exploiting vulnerabilities faster than manual identification
- Rapidly evolving attack patterns outpacing traditional defenses
- Heavy reliance on manual SOC analysis and static rule sets
- Delayed detection, high false-positive rates, and insufficient scalability
- Limited autonomous response capabilities against multi-stage attacks

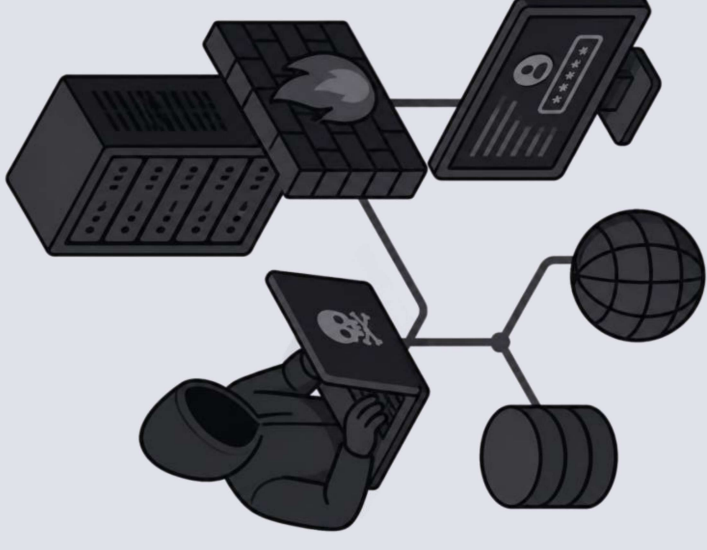
The Shift Required

Traditional Security

Manual Analysis → Delayed
Detection → Delayed
Response

AI-Driven Defense

Continuous Monitoring →
Intelligent Analysis →
Autonomous Response



Project Overview — AMCDS

Autonomous Multi-Agent Cyber Defense System

An AI-driven framework designed to simulate, detect, analyze, and autonomously respond to cyber threats in realtime network environments.



Multi-Agent Simulation

Coordinated attacker, defender, and SOC agents operating in a shared environment

Threat Detection

Continuous monitoring and real-time analysis of network activity



Dynamic Risk Assessment

GNN-powered scoring of evolving network risk states



Autonomous Response

LLM-assisted incident reporting and reinforcement-learning-driven containment

Project Objectives

- 1 Multi-Agent Simulation Environment**
Design autonomous attacker, defender, and SOC agents in a shared virtual network
- 2 MITRE ATT&CK Attack Scenarios**
Implement realistic staged attacks including reconnaissance, exploitation, and lateral movement
- 3 GNN-Based Risk Scoring**
Build AI-powered defense agents with continuous monitoring and graph-based threat scoring
- 4 Reinforcement Learning for Defense**
Optimize autonomous defensive decision-making through RL training
- 5 LLM Integration**
Leverage Large Language Models for threat interpretation and incident analysis
- 6 System Evaluation**
Assess effectiveness in improving incident response speed and network resilience

AMCDS System Architecture



Technology Stack



Programming & Infrastructure

- **Language:** Python
- **Backend:** FastAPI
- **Frontend:** React
- **Database:** PostgreSQL

AI / Machine Learning

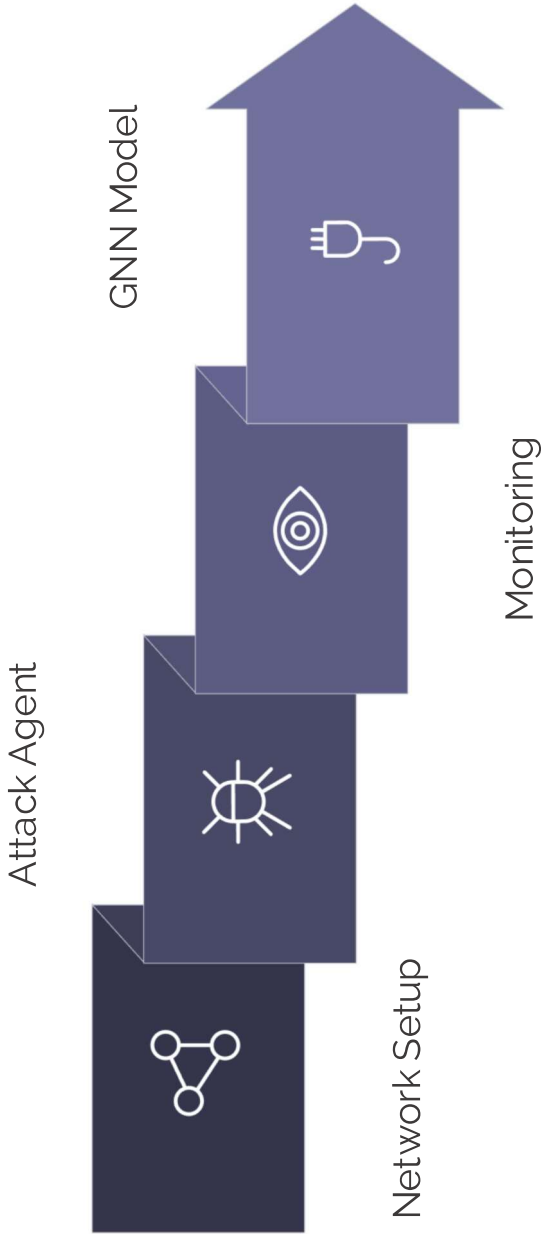
- **GNN:** Threat analysis and dynamic risk scoring
- **Reinforcement Learning:** Autonomous defensive decision-making
- **Large Language Models:** Threat interpretation and incident analysis

Security Framework

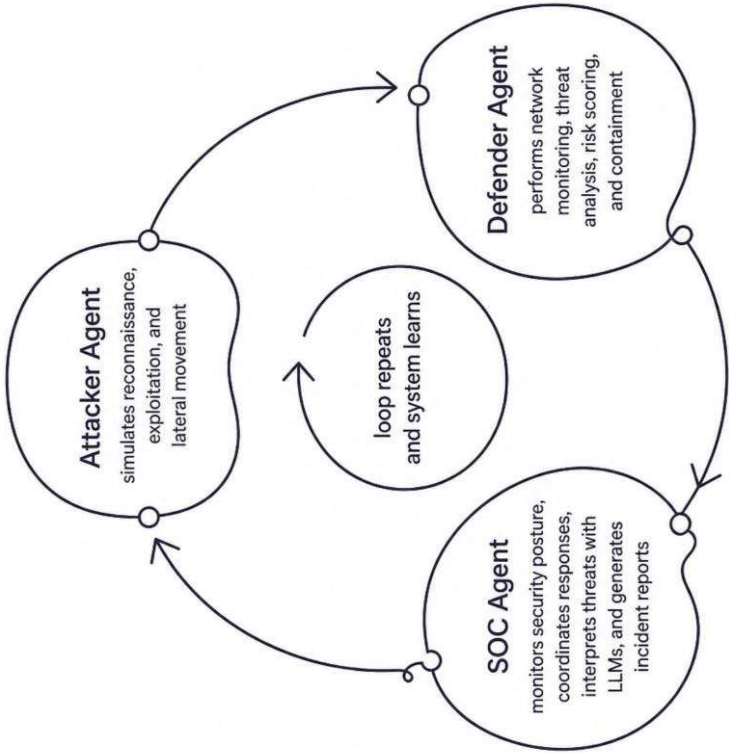
- **MITRE ATT&CK:** Attack simulation and threat modeling

Project Methodology

8-Stage Development Pipeline



Multi-Agent Cyber Defense Architecture



Agent Interaction Model

Agents operate in a continuous feedback loop — attackers generate activity, defenders detect and respond, SOC agents interpret and report, and the system learns and adapts over time.

Attacker Agent Simulates adversarial behavior: reconnaissance, exploitation, lateral movement	Defender Agent Monitors network, scores dynamic risk, executes automated containment	SOC Agent Interprets threats via LLMs, coordinates response, generates ind
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AI Technologies & Key Outcomes

AI Technologies Integrated

Integration Flow: GNN → Threat Understanding | RL → Defensive Decision | LLM → Incident Intelligence

Key Outcomes Achieved

- Automated Threat Detection
Real-time identification of multi-stage cyberattacks without manual intervention
- Dynamic Risk Assessment
Continuous GNN-driven scoring of evolving network threat states
- Improved Response Speed
Faster autonomous containment compared to manual SOC workflows
- Reduced Manual Intervention
Autonomous agent coordination minimizes dependency on human analysts
- Enhanced Network Resilience
System learns and adapts to novel attack patterns over simulation cycles

Graph Neural Networks (GNN)

Analyze host relationships and connections to calculate dynamic risk scores and map threat propagation paths

Reinforcement Learning (RL)

Trains defender agents to select optimal containment actions balancing threat neutralization against operational disruption

Large Language Models (LLMs)

Assist SOC agents in interpreting threats, summarizing incidents, and generating human-readable security reports